universität innsbruck

Introduction to GFlowNets

advaML Student Presentation

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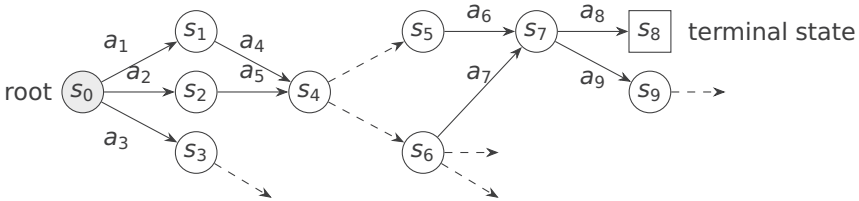
Recap and Notation

Directed Acylcic Graph (DAG)

- $G = (V, E)$ with nodes V and directed edges E
- no cycles

Markov Decision Process – $MDP(S, A, P_a, R_a)$

- state s_i ; action $a_i : s \mapsto s'$
- policy $\pi(s)$; reward $R(x)$
- trajectory τ



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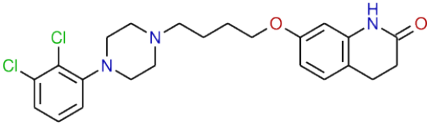
Motivation

Application: Design a drug/molecule to bind to a certain protein.

- Large and likely multimodal space.
  - Order of 10^{20} for single protein binding.
- We want to explore the space.
  - We might desire multiple possible designs to verify in testing.
  - Likely multimodal.
- Possibly not many positive examples on hand.



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Existing Methods and Shortcomings

Reinforcement Learning

Given a function $R(x)$, learn a $\pi(s)$ to maximize expected *reward*.

- often focuses probability mass to select *Trajectories*.
- multiple *trajectories* may reach the same *terminal state*.
- without *exploration* regularization, poor multimodal capturing.

MCMC

Generative Models

Existing Methods and Shortcomings

Reinforcement Learning

Given a function $R(x)$, learn a $\pi(s)$ to maximize expected *reward*.

MCMC

Given density $f(x)$, sample from simulated Markov chain with $p(x) \propto f(x)$.

- mixes slowly for high-dimensional or multimodal spaces.
- can get stuck at ’energy’ levels.
- high computational cost.

Generative Models

Existing Methods and Shortcomings

Reinforcement Learning
Given a function $R(x)$, learn a $\pi(s)$ to maximize expected *reward*.

MCMC
Given density $f(x)$, sample from simulated Markov chain with $p(x) \propto f(x)$.

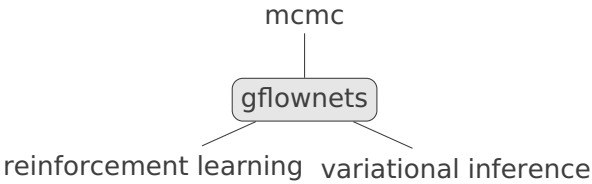
Generative Models
Given samples x_i of *target distribution* $X \sim p(x)$, learn a model/distribution that matches $p(x)$ with $q_\theta(x)$.

- Certain Training Objectives introduce bias.
- Few positive samples / we would like to use information of negative samples.

Approach

GFlowNets

- Given a *reward* function $R(x)$, sample *proportional* to $R(x)$.

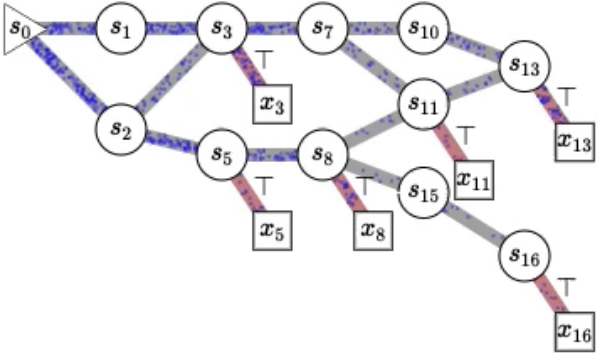


Goal

Learn a generative *Policy* $\pi(s)$ that samples *Actions* $a : s \mapsto s'$ proportional to a target reward $R(x)$.

Flow Matching

For every non-terminal state *s* the total **in-flow** equals the total **out-flow**.



[1]

Flow Matching

For every non-terminal state s the total **in-flow** equals the total **out-flow**.

Flow-Matching Constraint

$$\sum_{s,a:T(s,a)=s'} F(s,a) = R(s') + \sum_{a' \in A(s')} F(s',a')$$

with $R(s') = 0$ for non-terminal states and $F(x) = R(x) > 0$ for terminal nodes

Policy

Define Policy π that generates trajectories starting at s_0 by sampling actions $a \in A(s)$, such that:

Policy π

$$\pi(a|s) = \frac{F(s,a)}{F(s)}$$

- Where $f(s,a) > 0$ is the flow from state s through action a,
- $F(s) = R(s) + \sum_{a \in \mathcal{A}(s)} F(s,a)$
- $R(s) = 0$ for non-terminal nodes and $F(x) = R(x) > 0$ for terminal nodes

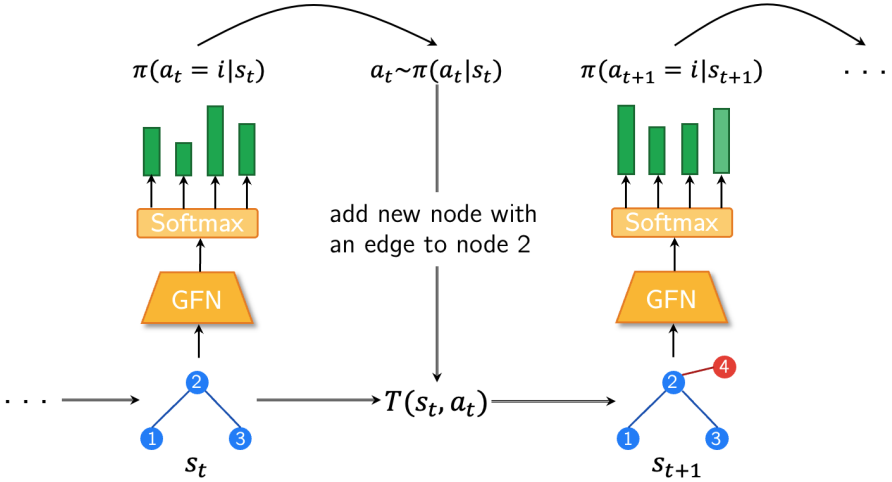
Proposition 2

- With the defined policy, the flow matching constraint, and having $\pi(s)$ denote the probability of visiting state s when starting at s_0 and following $\pi(\cdot|\cdot)$,

Proposition

$$
\begin{aligned}
&\text{(a)} \quad \pi(s) &&= \frac{F(s)}{F(s_0)} \\
&\text{(b)} \quad F(s_0) &&= \sum_{x \in \mathcal{X}} R(x) \\
&\text{(c)} \quad \pi(x) &&= \frac{R(x)}{\sum_{x' \in \mathcal{X}} R(x')}
\end{aligned}
$$

Structure of a GFlowNet



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Training Objective

Flow Matching:

$$\sum_{s,a:T(s,a)=s'} F(s,a) = R(s') + \sum_{a' \in A(s')} F(s',a')$$

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Loss:

$$\ell_{\mathrm{FM}}(s;\theta) = \Big(\sum_{s,a:T(s,a)=s'} F(s,a) - R(s') - \sum_{a' \in A(s')} F(s',a') \Big)^2.$$

Training Objective

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$$\tilde{\mathcal{L}}_\theta(\tau) = \sum_{s' \in \tau \neq s_0} \left(\sum_{s,a:T(s,a)=s'} F_\theta(s,a) - R(s') - \sum_{a' \in \mathcal{A}(s')} F_\theta(s',a') \right)^2$$

FM Training Objective

Flow Matching:

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Loss:

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$$\mathcal{L}_{\theta,\epsilon}(\tau) = \sum_{s' \in \tau \neq s_0} \left(\log \left[\epsilon + \sum_{s,a:T(s,a)=s'} \exp F_\theta^{\log}(s,a) \right] - \log \left[\epsilon + R(s') + \sum_{a' \in \mathcal{A}(s')} \exp F_\theta^{\log}(s',a') \right] \right)^2$$

Trajectory Balance

- State Flow $F(\cdot \mid s)$ instead of Edge Flow $F(s \to s')$
- Forward Policy $\pi_F(\cdot \mid s)$ & Backward Policy $\pi_B(\cdot \mid s)$

$$F(s \to s') = F(s)\pi_F(s' \mid s) = F(s')\pi_B(s \mid s')$$

$$\pi_F(s \mid s') = \frac{F(s \to s')}{\sum_{s^* : (s,s^*) \in \mathcal{A}} F(s \to s^*)}$$

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Loss:

$$\mathcal{L}_{\text{DB}}(s \to s') = \left(\log \frac{F_\theta(s)\, \pi_{F,\theta}(s' \mid s)}{F_\theta(s')\, \pi_{B,\theta}(s \mid s')} \right)^2$$

Trajectory Balance

- given a trajectory $\tau = (s_0 \to s_1 \to \ldots \to s_n = x)$
- match forward vs. backward flow for all states in τ:

$$\frac{F(s_0)\prod_{i=1}^{n}\pi_F(s_i \mid s_{i-1})}{R(x)\prod_{i=1}^{n}\pi_B(s_{i-1} \mid s_i)} = 1 \quad \iff \quad F(s_0)\,\pi_F(\tau) = R(x)\,\pi_B(\tau \mid x)$$

Trajectory Balance Loss

$$\mathcal{L}_{\text{TB}}(\tau) = \left(\log \frac{F_\theta(s_0)\pi_{F,\theta}(\tau)}{R(x)\pi_{B,\theta}(\tau \mid x)}\right)^2$$

Training process - overview

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Training process - overview

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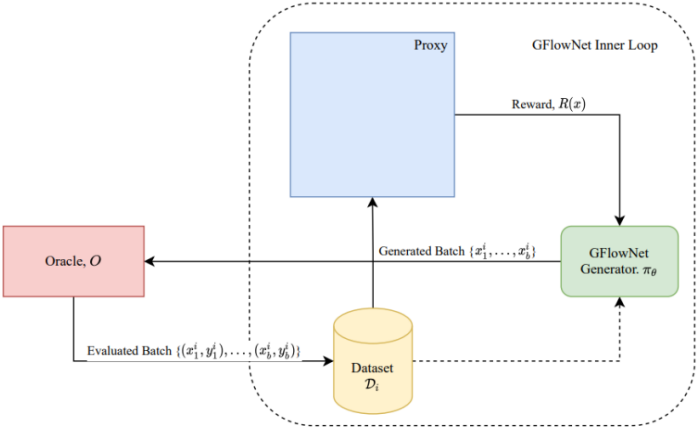
- Backpropagate loss and update net
- (Optional) Use a replay buffer to reuse diverse past trajectories
- Repeat until convergence

Multi-Round Setting



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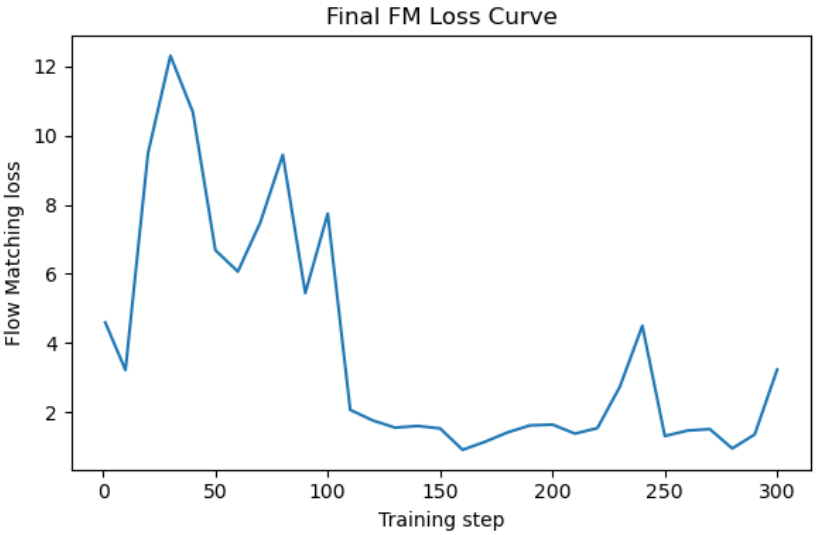
Demo

Tree Building

- 2 building blocks (branch, leaf)
- 3 positions per node
- possible actions every step
  - place building block x at free position n
  - end tree building
- defined max depth, max elements
- simple 2 layer linear network
- reward function
  - “energy” vs cost
  - reward per leaf, penalize per branch

Demo


Final FM Loss Curve
Flow Matching loss
Training step
12
10
8
6
4
2
0
50
100
150
200
250
300

References

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Generative flow networks, 2022.
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Yoshua Bengio, Kolya Malkin, and Moksh Jain.
The gflownet tutorial.
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Figure 2 referenced; accessed 23 May 2025.

Noah Hurmer and Ori Sofer.
Trajectory tree illustration — recreated from *Bengio et al.* using chatgpt o3.
Slide figure, 2025.
Figure redrawn with TikZ, adapted from the original diagram.

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In Kamalika Chaudhuri, Stefanie Jegelka, Le Song, Csaba Szepesvari, Gang Niu, and Sivan Sabato, editors, *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 9786–9801. PMLR, 17–23 Jul 2022.

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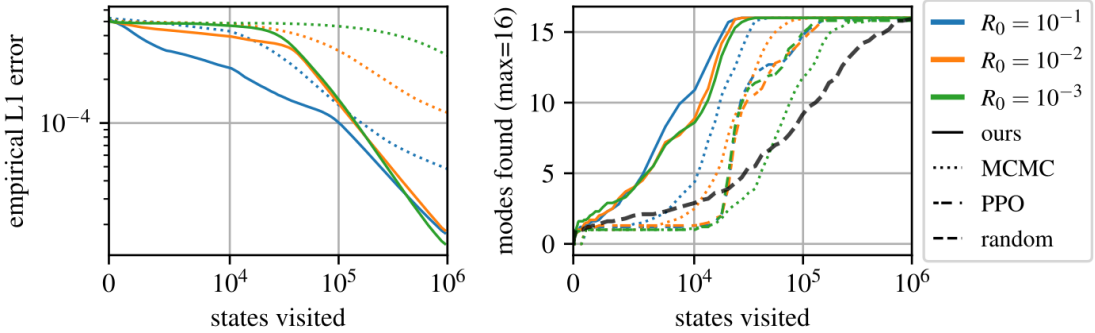
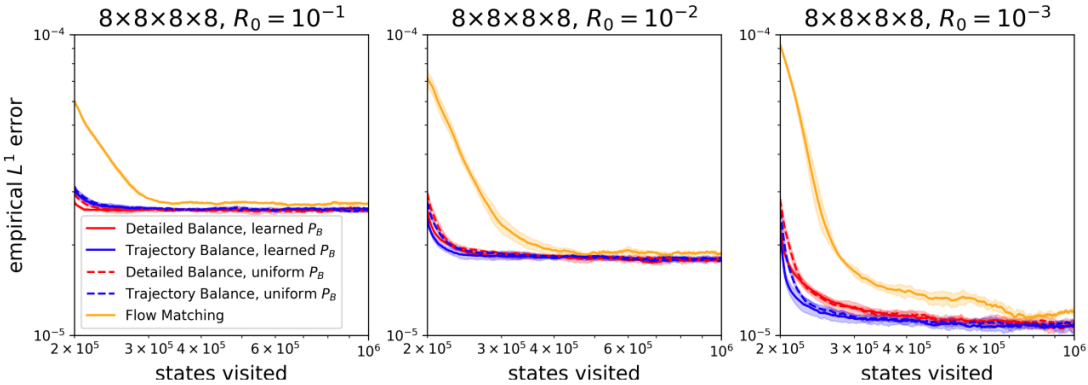
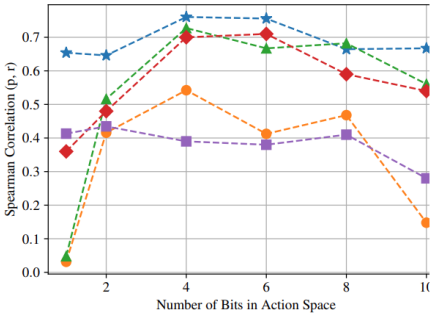
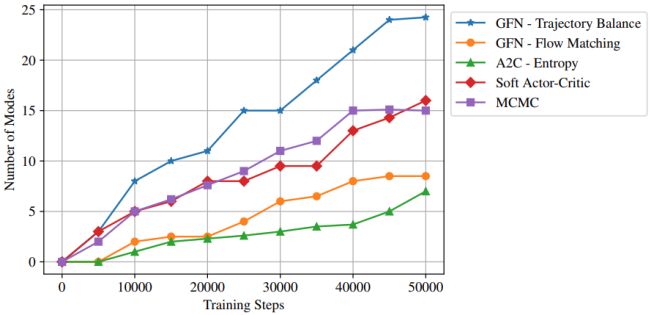
chemfig-examples: A few examples for the latex chemfig package.
https://github.com/jejrogers/chemfig-examples, 2025.
GitHub repository, accessed 23 May 2025.

Discussion


empirical L1 error
10^{-4}
0
10^4
10^5
10^6
states visited
modes found (max=16)
15
10
5
0
0
10^4
10^5
10^6
states visited
$R_0 = 10^{-1}$
$R_0 = 10^{-2}$
$R_0 = 10^{-3}$
ours
MCMC
PPO
random


8×8×8×8, $R_0 = 10^{-1}$
8×8×8×8, $R_0 = 10^{-2}$
8×8×8×8, $R_0 = 10^{-3}$
empirical L^1 error
10^{-4}
10^{-5}
2×10^5
3×10^5
4×10^5
6×10^5
10^6
states visited
Detailed Balance, learned P_B
Trajectory Balance, learned P_B
Detailed Balance, uniform P_B
Trajectory Balance, uniform P_B
Flow Matching


Spearman Correlation (ρ, r)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
2
4
6
8
10
Number of Bits in Action Space


Number of Modes
0
5
10
15
20
25
0
10000
20000
30000
40000
50000
Training Steps
GFN - Trajectory Balance
GFN - Flow Matching
A2C - Entropy
Soft Actor-Critic
MCMC